Q1: For some event A with $P(A) = 0.1$ then $P(\bar{A}|\Omega) =$

A) 0.1 B) 0.8 C) 0.2 D) 0.9 E) None

Q2: Let X be a random variable with mean $a > 0$. If $E(X^2) = 2a^2$ and $Var(X) = 1$. Then a equals

A) 0 B) 1 C) 2 D) 3 E) None

Q3: For some independent events A, B with $P(A) = P(B)$ and $P(A \cap B) = \frac{1}{16}$ Then $P(A) =$

A) 0.2 B) 0.1 C) 0.25 D) 0.5 E) None

Q4: For $x > 0$ we have $u(-x) =$

A) 1 B) 0 C) *undefined* D) -1 E) None

Q5: For $R_{XY} = \{(0,0), (1,2)\}$ if $E(XY) = 0.2$ Then $f(0,0) =$

A) 0.1 B) 0.2 C) 0.8 D) 0.9 E) None

Q6: For $R_{XY} = \{(0,0), (1,1)\}$ if $f(0,0) = f(1,1)$ then $E(Xe^Y) =$

A) $e/2$ B) 0 C) $2e$ D) e E) None

Q7: $\int_{-\infty}^{\infty} \sin(x)\, \delta(x + \frac{\pi}{2})\, dx =$

A) 1 B) 8 C) -8 D) -1 E) None

Q8: For the joint PDF $f(x,y) = 1$ when $0 < x < 1$, $0 < y < 1$ and zero otherwise. We have $f_Y(y) =$

A) 1 B) 0.5 C) 0.25 D) 2 E) None

Q9: For continuous random variable X with CDF $F(x)$. If $F(0) = 0$, $F(1) = 0.2$ and $F(2) = 1$ Then $P(0 < X < 1) + P(0 < X < 2) =$

A) 0.3 B) 1.2 C) 1.3 D) 0.2 E) None

Q10: For random variables X, Y. If $Cov(1 - X, 1 - Y) = Corr(-X, -Y)$ then $\sigma_X \sigma_Y =$

A) 3 B) 2 C) 1 D) 0.5 E) none

Q11: Let X, Y be two independent random variables with $E(X) = E(XY) \neq 0$ Then $E(Y) =$

A) 1 B) 9 C) 2 D) 16 E) None

Q12: A discrete random variable X taking two values $0, -1$ with equal probability, then its generalized PDF $P(X = x)$ equals

A) $0.5\,\delta(x) + 0.5\,\delta(x-1)$ B) $\delta(x) + \delta(x+1)$
C) $\delta(x) + \delta(x-1)$ D) $0.5\,\delta(x) + 0.5\,\delta(x+1)$ E) None

Q13: How many possible outcomes are there when two coins are tossed three times?

A) 64 B) 32 C) 16 D) 8 E) None

Q1: $P(\bar{A} \mid \Omega) = P(\bar{A}) = 1 - 0.1 = 0.9$

Q2: $EX = a$, $EX^2 = 2a^2$, $Var(X) = 1$

$Var(X) = 1 = EX^2 - (EX)^2 = 2a^2 - a^2 = a^2$

(given that $a > 0$) $a = 1$.

Q3: $p(A \cap B) = p(A) \cdot p(B) = p^2(A) = 1/16$

$p(A) = +1/4$

Q4: $u(-x) = \begin{cases} 0, & x > 0 \\ 1, & x \leq 0 \end{cases}$

Q5: discrete r.v so

$EXY = \sum\sum xy\, f(x,y)$

$0.2 = 0 \cdot f(0,0) + (1)(2) f(1,2)$

$0.2 = 2 f(1,2) \longrightarrow f(1,2) = 0.1$

$1 = f(0,0) + f(1,2) \longrightarrow f(0,0) = 0.9$

Q6: discrete r.v

$f(1,1) = f(0,0) \longrightarrow 2f(1,1) = 1 \rightarrow f(1,1) = 0.5$

$EXe^Y = 0e^0 f(0,0) + (1)e^1 f(1,1) = e/2$

Q7: $\int_{-\infty}^{\infty} \sin(x)\, \delta(x-(-\pi/2))\, dx = \sin(-\pi/2) = -1$

Q8: $f_Y(y) = \int_{R_X} f(x,y)dx = \int_0^1 dx = x\Big]_0^1 = 1$

Q9: $p(0<X<1) = F(1) - F(0) = 0.2$

$p(0<X\leq 2) = F(2) - F(0) = \frac{1}{0.2}$ +

Q10: $Cov(1-X, 1-Y) = (-1)(-1)Cov(X,Y) = Cov(X,Y)$

$Corr(-X, -Y) = Corr(X,Y)$

نفس یا متفاوت

* $Corr(X,Y) \neq \frac{Cov(X,Y)}{\sigma_X \sigma_Y}$

$\sigma_X \sigma_Y = \frac{Cov(X,Y)}{Corr(X,Y)} = -1$

Q11: Keyword: indep. r.vs

$Cov(X,Y) = 0 = EXY - (EX)(EY)$

$(EX)(EY) = EX \longrightarrow EY = 1$

Q12: $X = \{-1, 0\}$, equal prob. $p(x=1) = p(x=2) = 1/2$

$P(X=x) = 0.5\,\delta(x) + 0.5\,\delta(x-(-1))$

$= 0.5\,\delta(x) + 0.5\,\delta(x+1)$

Q13: 2 coins tossed 3 times $\equiv$ 1 coin tossed 6 times

$|\Omega| = 2^6 = 64$

Q1: For some event A with $P(A) = 0.9$ then $P(\bar{A}|\Omega) =$

A) 0.1 B) 0.8 C) 0.2 D) 0.9 E) None

Q2: Let X be a random variable with mean $a > 0$. if $E(X^2) = 2a^2$ and $Var(X^2) = 4$. Then a equals

A) 0 B) 1 C) 2 D) 3 E) None

Q3: For some independent events A, B with $P(A) = P(B)$ and $P(A \cap B) = \frac{1}{4}$

Then $P(A) =$

A) 0.2 B) 0.1 C) 0.25 D) 0.5 E) None

Q4: For $x < 0$ we have $u(-x) =$

A) 1 B) 0 C) $undefined$ D) -1 E) None

Q5: For $R_{XY} = \{(0,0), (1,2)\}$, if $E(XY) = 1.6$ Then $f(0,0) =$

A) 0.1 B) 0.2 C) 0.8 D) 0.9 E) None

Q6: For $R_{XY} = \{(0,0), (1,1)\}$, if $f(0,0) = f(1,1)$ then $E(2Xe^Y) =$

A) $e/2$ B) 0 C) $2e$ D) e E) None

Q7: $\int_{-\infty}^{\infty} \sin(x)\, \delta(x + \frac{\pi}{2})\, dx =$

A) -1 B) 8 C) -8 D) 1 E) None

Q8: For the joint PDF $f(x,y) = 1$ when $0 < x < 1$, $0 < y < 1$ and zero otherwise. We have $f_X(x) =$

A) 0.25 B) 0.5 C) 1 D) 2 E) None

Q9: For continuous random variable X with CDF $F(x)$. If $F(0) = 0$, $F(1) = 0.2$ and $F(2) = 1$ Then $P(0 < X < 1) + P(0 < X < 2) =$

A) 0.3 B) 0.2 C) 1.3 D) 1.2 E) None

Q10: For random variables X, Y. If $Cov(1 - X, 1 - Y) = Corr(-X, -Y)$ then $\sigma_X \sigma_Y =$

A) 3 B) 1 C) 2 D) 0.5 E) none

Q11: Let X, Y be two independent random variables with $E(X) = E(XY) \neq 0$ Then $E(Y) =$

A) 16 B) 9 C) 2 D) 1 E) None

Q12: A discrete random variable X taking two values 0,1 with equal probability, then its generalized PDF $P(X = x)$ equals

A) $0.5\,\delta(x) + 0.5\,\delta(x-1)$ B) $\delta(x) + \delta(x+1)$
C) $\delta(x) + \delta(x-1)$ D) $0.5\,\delta(x) + 0.5\,\delta(x+1)$ E) None

Q13: How many possible outcomes are there when two coins are tossed three times?

A) 8 B) 32 C) 16 D) 64 E) None

Q1: $P(\bar{A}|\Omega) = 1 - P(A|\Omega) = 1 - P(A) = 0.1$

Q2: $EX = a$, $EX^2 = 2a^2$, $Var(X) = 4$

$$Var(X) = 4 = EX^2 - (EX)^2 = 2a^2 - a^2 - a^2$$

$$a^2 = 4 \longrightarrow a = 2$$

Q3: $P(A \cap B) = P(A) \cdot P(B) = P^2(A) = 1/4$

$$P(A) = 1/2$$

Q4: $u(-x) = \begin{cases} 0, & x > 0 \\ 1, & x \leq 0 \end{cases}$

Q5: $EXY = \sum\sum xy\, f(x,y)$

$$1.6 = 0\, f(0,0) + (1)(2) f(1,2)$$

$$1.6 = 2f(1,2) \longrightarrow f(1,2) = 0.8$$

$$f(0,0) + f(1,2) = 1 \longrightarrow f(0,0) = 0.2$$

Q6: $f(0,0) = f(1,1) \longrightarrow f(1,1) = 0.5$

$$2EXe^{y} = 0 \cdot e^{0} f(0,0) + 1\, e^{1} f(1,1) = 2 \times \frac{1}{2} e = e$$

Q7: $$\int_{-\infty}^{\infty} \sin(x)\, \delta\left(x - \left(-\tfrac{\pi}{2}\right)\right) dx = \sin\left(-\tfrac{\pi}{2}\right) = -1$$

Q8: $$f_x(x) = \int_{R_y} f(x,y)\, dy = \int_0^1 dy = y\Big]_0^1 = 1$$

Q9: $P(0<X<1) = F(1) - F(0) = 0.2$

$P(0<X<2) = F(2) - F(0) = \dfrac{1}{1.2}$ +

Q10: $Cov(1-X, 1-Y) = (-1)(-1)\, Cov(X,Y) = Cov(X,Y)$

$Corr(-X,-Y) = Corr(X,Y)$

نفس الإشارات

$$Corr(X,Y) \not= \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

$$\sigma_x \sigma_y = \frac{Cov(X,Y)}{Corr(X,Y)} = 1$$

Q11: keyword: indep. r.v , $EX \neq 0$

$$Cov(X,Y) = 0 = EXY - (EX)(EY)$$

$$0 = EX(1 - EY) \longrightarrow EY = 1$$

Q12: $X = \{0, 1\}$, equil prob. $p(x=0) = p(x=1) = 1/2$

$$P(X = x) = 0.5\,\delta(x) + 0.5\,\delta(x-1)$$

Q13:

tossing		tossing
2 coins	=	a coin
3 times		6 times

$$|\Omega| = 2^6 = 64$$

Q14 : Given the CDF

$$F(x)=\begin{cases}0 & x<1\\ 0.2 & 1\le x<2\\ 0.9 & 2\le x<3\\ 1 & 3\le x\end{cases}$$

Find $P(X=2)$ and $P(X=3)$

b.d.f =

x	Px
1	0.2
2	0.7
3	0.1

$P(x=2)=0.7$

$P(x=3)=0.1$

Q15 : For the joint PDF $f(x,y)=u(x)u(1-x)u(y)e^{-y}$. find $f_x(x)$

$$f_x(x)=\int_{-\infty}^{\infty} u(x)\,u(1-x)\,u(y)\,e^{-y}\,.dy$$

$$=\int_{0}^{\infty} u(x)\,u(1-x)\,e^{-y} \rightarrow u(x)\,u(1-x)\,\frac{e^{-y}}{-1}\Big|_{0}^{\infty}$$

$$=u\times u(1-x)\,[0--1]=\boxed{u(x)\,u(1-x)}\rightarrow f_x(x)$$

- الأسئلة المقالية :

Q14 :

if $F(x) = \begin{cases} 0, & x > 1 \\ 0.2, & 1 \leq x < 2 \\ 0.9, & 2 \leq x < 3 \\ 1, & 3 \geq x \end{cases}$

find $p(x=2)$, $p(x=3)$:

Sol :

$p(x=1) = 0.2$

$0.9 = p(x=1) + p(x=2)$

$0.9 - 0.2 = p(x=2) = 0.7$

$p(x=1) + p(x=2) + p(x=3) = 1$

$1 - 0.7 - 0.2 = p(x=3) = 0.1$

$p(x=2) = 0.7$, $p(x=3) = 0.1$

Q15: if $f(x,y) = u(x)\,u(1-x)\,u(y)\,e^{-y}$

find $f_x(x)$:

Sol:

$$f_x(x) = \int_{R_y} f(x,y)\,dy$$

$$= u(x)\,u(1-x) \int_{-\infty}^{\infty} u(y)\,e^{-y}$$

$$= u(x)\,u(1-x) \int_{0}^{\infty} 1 \cdot e^{-y}\,dy$$

$$= u(x)\,u(1-x) \left(-e^{-y} \Big]_0^{\infty} \right)$$

$$= u(x)\,u(1-x) \left(e^{-y} \Big]_{\infty}^{0} \right)$$

$$= u(x)\,u(1-x) \left(1 - \lim_{t \to \infty} e^{-t} \right)$$

$$f_x(x) = u(x)\,u(1-x)$$